Tactix: DSR proof-state boundary and producer-source audit

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Current result. The full repaired-source proofless Rank 1–50 diagnostic remains terminal 50/50, with 25 independently checked SAT models and 25 matching UNSAT verdicts. This is verdict/model parity against the historically checked automatic-proof reference, not a proof certificate for its proofless UNSAT runs. The five SAT cases lost under the old source are recovered on the repaired binary in both automatic-proof and proofless targeted arrays. The current-source full automatic-proof Rank/B28+C5 gate remains unsubmitted while the native DSR proof failure is resolved.

General DSR boundary. The published AM465 internal Proof API D2F handoff was rebuilt against the current serializer, whose source is unchanged since the AM472 batch repair. Job `1729601` reproduced byte-identical DSR proofs for both bounded seeds. The input-root-reason case rejects in both no-option `dsr-trim` directions; the deleted-unit case rejects forward, and its backward elaboration is rejected by Trestle (`1729708`). A separate control `1729623` inserted the root-unit materialization that the production kernel would perform before deleting the input reason. The general split transaction appeared, but both checker directions still rejected the later witnessed line. The input's own unit propagation had already made the deleted clause a checker root reason; moving deletion ahead of newly emitted units cannot undo that input state. This identifies a proof-state boundary, not a new one-case serializer patch.

The internal program is not itself a public-rule failure. A current-binary control using public Probe, BCE, and explicit X7 on the same CNF (`1729650`) took a different path: BCE declined and X7 reached UNSAT. Its complete proof verified in both no-option DSR directions and Trestle (`1729665`). Current-proof-source Opus AM473 still owns the broader public-policy fuzz handoff. No public X7 guard, checker option, or plan section 5 route was changed. Earlier fixed-corpus controls show that `-delete-implied-units` is not a monotone checker relaxation, so it cannot be silently substituted for the canonical no-option contract.

Winner producer-source audit. A read-only scan of the pinned SC24 Kissat, AE-Kissat-MAB, and CaDiCaL 3.0.0 producer call sites maps nonincremental learned/unit/empty additions, strengthening, blocked/covered/autarky deletions, equivalence rewrites, fresh-variable Factor RAT clauses, and model-extension records to Tactix's declared RUP/RAT/delete/reconstruction action catalog. This extends the prior emitter-interface audit to direct source producers; it does not verify every public Tactix instance or certificate, or scheduling parity. CaDiCaL's incremental query protocol remains deferred. The separate active winner gap is general live pseudo-Boolean arithmetic, redundancy and dominance in listed LymphoSAT paths and Satsuma's VeriPB mode; Tactix's PB path is startup-refutation-only. No general PB action was added here.

Next decision. Obtain AM473's bounded public-policy result and design a general checker-visible proof-state or certificate-construction repair that preserves legal X7 instances. Then rerun current-source Opus fuzz, full Debug, independent proof/model controls, Rank/B28+C5, WV100, and WV2-89 gates. Only after N1–N4 and learning readiness are verified should the 1,166-instance Tier D run be submitted at 5,000 seconds, 4 CPUs and 32 GiB on `cpu-dense-hwulab-q`. It has not been submitted.

Evidence. `benchmarks/reports/2026-09-26-am476-d2f-current-source-boundary.md`; `benchmarks/reports/2026-09-26-am477-cadical-kissat-proof-producer-map.md`; `benchmarks/reports/2026-09-26-am475-five-sat-recovery.md`; `benchmarks/reports/2026-09-25-am407-archived-dsr-contract-crosscheck.md`.